



PERTH MODERN SCHOOL
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Test Two

Semester One 2018 ***UNIT 1 METHODS***

Calculator Free 40 minutes /40 marks

Formula Sheet is permitted

Name:

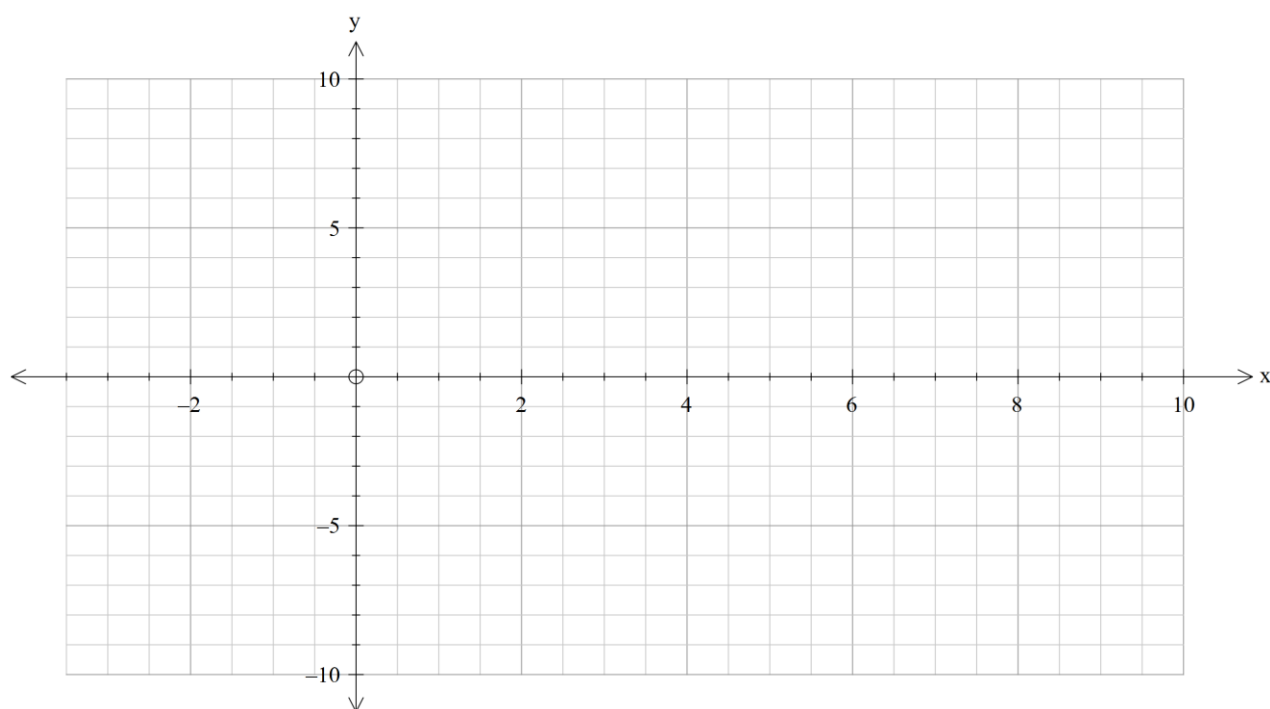
Place a tick in the box next to your Mathematics teachers name:

Mr Strain	☐
Ms Sindel	☐
Mrs Rimando	☐
Mr Gannon	☐
Mr Young	☐
Mrs Flynn	☐
Ms Ensly	☐

Question 1**(4, 2, 2 = 8 marks)**

Given $(y + 1)^2 = 2x - 1$,

i) sketch the graph of the equation.

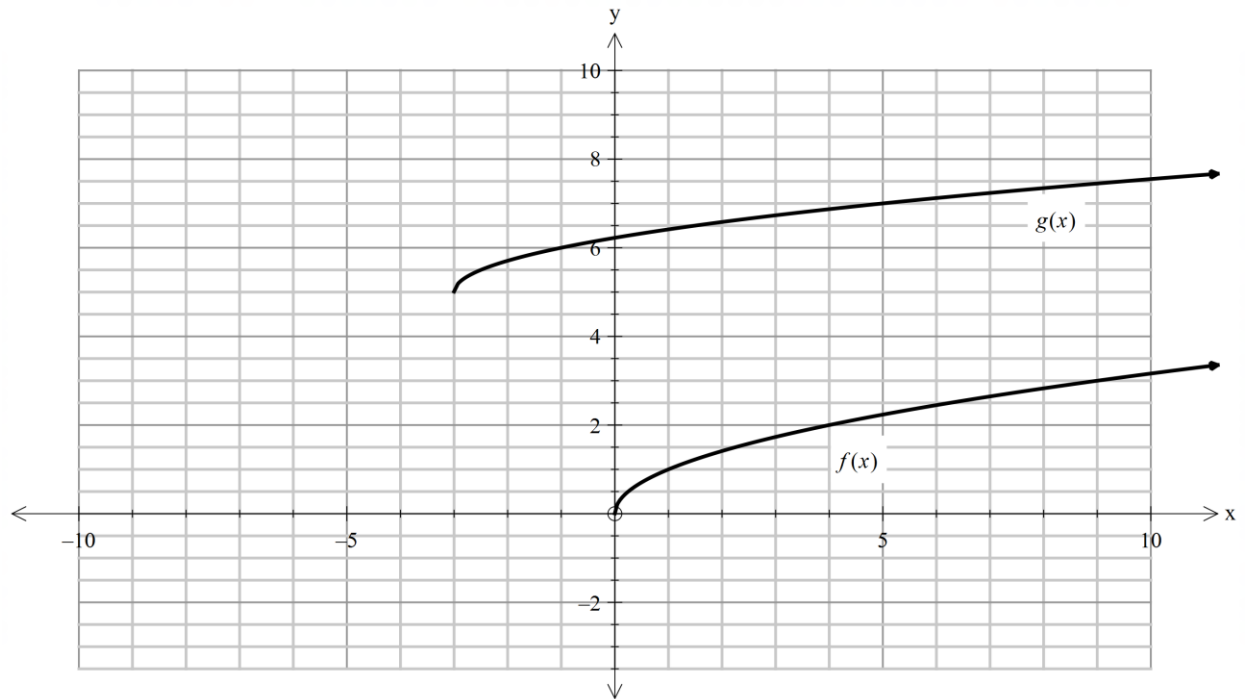


ii) state its domain and range.

iii) Is this graph a function? Justify.

Question 2**(4, 3 = 7 marks)**

The function $f(x) = \sqrt{x}$ undergoes several transformations that result to $g(x)$ as represented in the graphs below.



i) Identify the order of transformations that would transform $f(x)$ to $g(x)$.

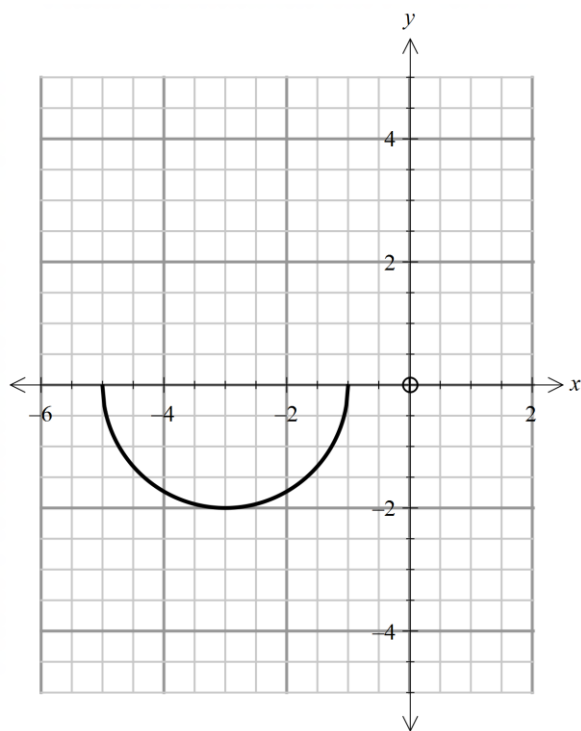
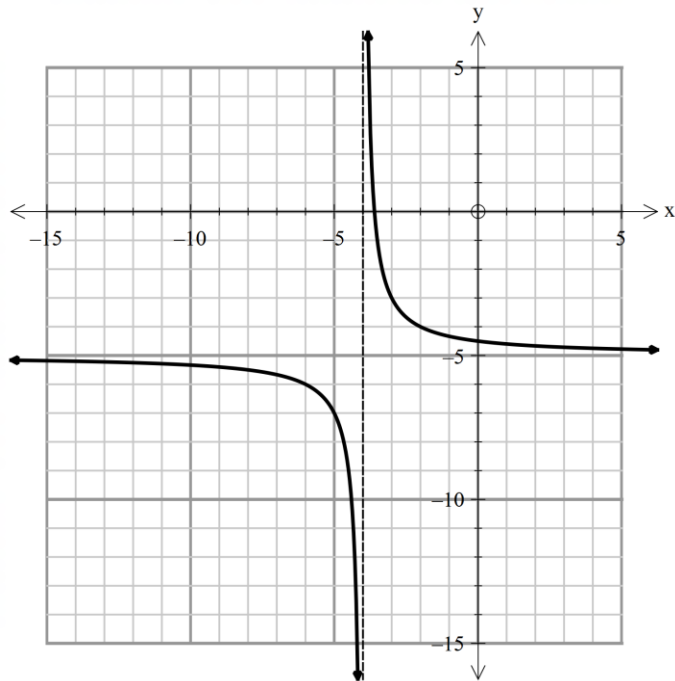
ii) Write the equation of the resulting function $g(x)$.

Question 4

(3, 3 = 6 marks)

Write the equations of the following graphs:

i)



ii)

Question 5**(2, 2, 2 = 6 marks)**

A pebble is thrown vertically upwards. It has an initial speed of u metres per second. The pebble reaches a maximum height of h metres before falling vertically downwards. It is known that h is directly proportional to u^2 . When the pebble is thrown with an initial speed of 10 m/s , it reaches a maximum height of 5 meters.

- i) Write an equation that models this relationship.

- ii) Calculate the maximum height reached when the pebble is thrown with an initial speed of 12 m/s .

- iii) Find the initial speed of the pebble if the maximum height reached is 16 meters. Write your answer as an exact value.

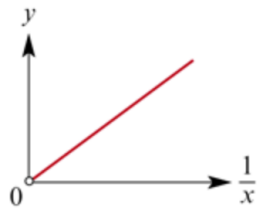
Question 6**(4 marks)**

State whether the relationships given below model a direct variation, inverse variation or neither.

i) $y = 7x - 2$

ii) The number of hours to finish a job and the number of workers.

iii)



iv)

x	1	2	3
y	5	2.5	1

Question 7

(1, 4 =5 marks)

There are 100 people in an evacuation centre. Supplies for food are being rationed and their supply will last for 20 days.

- i) Write a variation statement that relates the number of days for food supplies f , to the number of people, p .
- ii) Suppose 25 more people are evacuated in the same centre. At the very latest, on what day should their supplies be replenished?

End of test